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Modulated Brightness Adjustment for Reflective Displays

Abstract

An electronic device having a display is provided. The device may include a projector that provides image light to a waveguide for display at an eye box. The projector may include light sources that provide illumination light to a reflective display panel. The panel may generate the image light by modulating image data onto the illumination light. To mitigate postcard artifacts at the eye box during low light conditions, the projector may perform hierarchical dimming that includes analog dimming, modulation-based dimming, and digital dimming as luminance decreases. The modulation-based dimming may include reducing a bit depth of a display modulation sequence in the image data and compressing the display modulation sequence over time. A display modulation sequence may be time-stretched to match the brightness of a preceding display modulation sequence to prevent flicker.

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Background/Summary

[0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/558,452, filed Feb. 27, 2024, which is hereby incorporated by reference herein in its entirety.

FIELD

[0002] This relates generally to electronic devices, including electronic devices with displays.

BACKGROUND

[0003] Electronic devices often include displays that display images. The displays include optics that redirect the images for view by a user. It can be challenging to design electronic devices with displays that display high quality images across a variety of usage contexts.

[0004] For example, consider a scenario in which a user is operating an electronic device in a dark environment. If care is not taken, light leakage and other non-idealities in the optics can minimize contrast of the images and/or can produce visible artifacts that can be distracting to the user. It is within this context that the embodiments herein arise.

SUMMARY

[0005] An aspect of the disclosure provides an electronic device. The electronic device can include one or more light sources configured to generate illumination light based on a drive current. The electronic device can include a reflective display panel configured to receive image data and configured to generate image light by reflecting the illumination light based on the image data. The electronic device can include a waveguide configured to direct the image light towards an eye box. The electronic device can include one or more processors configured to adjust the image light between a first brightness and a second brightness using the drive current, the second brightness being lower than the first brightness. The one or more processors can be configured to adjust the image light between the second brightness and a third brightness by adjusting bit planes of a display modulation sequence in the image data, the third brightness being lower than the second brightness.

[0006] An aspect of the disclosure provides an electronic device. The electronic device can include one or more light sources configured to generate illumination light. The electronic device can include a reflective display panel configured to receive image data and configured to generate image light by reflecting the illumination light based on the image data, wherein the image data implements a display modulation sequence. The electronic device can include a waveguide configured to direct the image light towards an eye box. The electronic device can include one or more processors configured to dim the image light by at least reducing a bit depth of the display modulation sequence.

[0007] An aspect of the disclosure provides a method of operating an electronic device. The method can include generating, using one or more light sources, illumination light. The method can include generating, using a reflective display panel, image light based on the illumination light. The method can include directing, using a waveguide, the image light towards an eye box. The method can include dimming, using one or more processors, the image light. Dimming the image light can include driving, at a first time, the reflective display panel using a first sequence of bit planes for a first duration, and driving, at a second time after the first time, the reflective display panel using the first sequence of bit planes for a second duration less than the first duration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a diagram of an illustrative system having a display in accordance with some embodiments.

[0009] FIG. **2** is a top view of an illustrative optical system for a display having a waveguide that receives light from a projector having a reflective display panel in accordance with some embodiments.

[0010] FIG. **3** is a top view of an illustrative projector having a reflective display panel in accordance with some embodiments.

[0011] FIG. **4** is a timing diagram of an illustrative display modulation sequence that may be used by a projector having a reflective display panel to produce light that is provided to a waveguide in accordance with some embodiments.

[0012] FIG. **5** is a flow chart of illustrative operations involved in adjusting the brightness of a projector having a reflective display panel in accordance with some embodiments.

[0013] FIG. **6** is a flow chart of illustrative operations involved in performing hierarchical brightness adjustments using a projector having a reflective display panel in accordance with some embodiments.

[0014] FIG. **7** is a diagram showing how an illustrative reflective display panel may perform modulation-based dimming in accordance with some embodiments.

[0015] FIG. **8** is a flow chart of illustrative operations involved in performing modulation-based dimming using a reflective display panel in accordance with some embodiments.

DETAILED DESCRIPTION

[0016] An illustrative system **10** having a device with one or more near-eye display systems is shown in FIG. **1**. System **10** may be an electronic device such as a head-mounted device (HMD) having one or more near-eye displays such as displays **14** mounted within support structure (housing) **20**. System **10** may be, for example, a virtual reality, mixed reality, and/or augmented reality headset, helmet, goggles, or glasses. Displays **14** may include one or more display projectors such as projectors **14A** and one or more optical systems such as optical systems **14B**. Projectors **14A** may be mounted in a support structure such as support structure **20**. Each projector **14A** may output light **22** (sometimes referred to herein as image light **22**) that is redirected towards a user's eyes at eye box **24** using an associated one of optical systems **14B**.

[0017] Support structure **20** may have the shape of a pair of eyeglasses (e.g., supporting frames), may form a housing having a helmet shape, or may have other configurations to help in mounting and securing the components of displays **14** on the head or near the eye of a user. Support structure **20** may, for example, may include portions (e.g., head-mounted support structures) formed from fabric, polymer, metal, and/or other material. Support structure **20** may include a strap or other head-mounted support structures to help support system **10** on a user's head. Support structure **20** may include a main housing portion that supports electronic components and/or optical components such as optical systems **14B**. Support structure **20** may include temple portions that extend from opposing sides of the main housing portion (e.g., for placement over or on the ears of the user). The temple portions may be rotatable relative to the main housing portion or may be at a fixed orientation relative to the main housing portion. Projectors **14A** may be disposed in the temple portions if desired.

[0018] The operation of system **10** may be controlled using control circuitry **16**. Control circuitry **16** may include storage and processing circuitry for controlling the operation of system **10**. Control circuitry **16** may include storage such as hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Control circuitry **16** may include

processing circuitry such as one or more processors (e.g., microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, central processing units, etc.). Software code (instructions) may be stored on storage in control circuitry **16** and executed by processing circuitry in control circuitry **16** to implement operations for system **10** (e.g., data gathering operations, operations involving the adjustment of components using control signals, image rendering operations to produce image content to be displayed for a user, etc.).

[0019] System **10** may include input-output circuitry such as input-output devices **12**. Input-output devices **12** may be used to allow data to be received by system **10** from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, or other electrical equipment) and to allow a user to provide system **10** with user input. Input-output devices **12** may also be used to gather information on the environment in which system **10** (e.g., a head-mounted device) is operating. Output components in devices **12** may allow system **10** to provide a user with output and may be used to communicate with external electrical equipment. Input-output devices **12** may include sensors and other components **18**. If desired, components **18** may include gaze tracking sensors that gather gaze image data from a user's eye at eye box **24** to track the direction of the user's gaze in real time.

[0020] Components **18** may include one or more sensors. Sensors in components **18** may include force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors such as a touch sensor that forms a button, trackpad, or other input device), and other sensors. If desired, the sensors in components **18** may include optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, optical touch sensors, optical proximity sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, image sensors (e.g., cameras), fingerprint sensors, iris scanning sensors, retinal scanning sensors, and other biometric sensors, temperature sensors, sensors for measuring three-dimensional non-contact gestures (“air gestures”), pressure sensors, sensors for detecting position, orientation, and/or motion of system **10** and/or information about a pose of a user's head (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or inertial measurement units that contain some or all of these sensors), health sensors such as blood oxygen sensors, heart rate sensors, blood flow sensors, and/or other health sensors, radio-frequency sensors, three-dimensional camera systems such as depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices that capture three-dimensional images) and/or optical sensors such as self-mixing sensors and light detection and ranging (lidar) sensors that gather time-of-flight measurements (e.g., time-of-flight cameras), humidity sensors, moisture sensors, gaze tracking sensors, electromyography sensors to sense muscle activation, facial sensors, and/or other sensors. In some arrangements, system **10** may use sensors in components **18** and/or other input-output devices to gather user input. For example, buttons may be used to gather button press input, touch sensors overlapping displays can be used for gathering user touch screen input, touch pads may be used in gathering touch input, microphones may be used for gathering audio input (e.g., voice commands), accelerometers may be used in monitoring when a finger contacts an input surface and may therefore be used to gather finger press input, etc.

[0021] Projectors **14A** (sometimes referred to herein as display engines **14A**, light engines **14A**, or display modules **14A**) may include reflective displays (e.g., displays with a light source that produces illumination light that reflects off of a reflective display panel to produce image light, such as liquid crystal on silicon (LCOS) displays, digital-micromirror device (DMD) displays, or other spatial light modulators), emissive displays (e.g., micro-light-emitting diode (uLED) displays, organic light-emitting diode (OLED) displays, laser-based displays, etc.), or displays of other types. Light sources in projectors **14A** may include uLEDs, OLEDs, LEDs, lasers, combinations of these, or any other desired light-emitting components. Implementations in which projectors **14A** include

reflective displays are described herein as an example.

[0022] Optical systems **14B** may form lenses that allow a viewer (see, e.g., a viewer's eyes at eye box **24**) to view images on display(s) **14**. There may be two optical systems **14B** (e.g., for forming left and right lenses) associated with respective left and right eyes of the user. A single display **14** may produce images for both eyes or a pair of displays **14** may be used to display images. In configurations with multiple near-eye displays (e.g., left and right eye displays), the focal length and positions of the lenses formed by components in optical system **14B** may be selected so that any gap present between the displays will not be visible to a user (e.g., so that the images of the left and right displays overlap or merge seamlessly).

[0023] If desired, optical system **14B** may contain components (e.g., an optical combiner, etc.) that allow real-world image light from real-world images or objects **25** to be combined optically with virtual (computer-generated) images such as virtual images in image light **22**. In this type of system, which is sometimes referred to as an augmented reality system, a user of system **10** may view both real-world objects and computer-generated content (e.g., virtual objects) that is overlaid on top of the real-world objects. Camera-based augmented reality systems may also be used in system **10** (e.g., in an arrangement in which a camera captures real-world images of object **25** and this content is digitally merged with virtual content at optical system **14B**).

[0024] System **10** may, if desired, include wireless circuitry and/or other circuitry to support communications with a computer or other external equipment (e.g., a computer that supplies display **14** with image content). The wireless circuitry may include antennas, radio-frequency transceiver circuitry, and other wireless communications circuitry and/or wired communications circuitry. The wireless circuitry may support bidirectional wireless communications between system **10** and external equipment (e.g., a companion device such as a computer, cellular telephone, or other electronic device, an accessory such as a point device or a controller, computer stylus, or other input device, speakers or other output devices, etc.) over one or more wireless links. During operation, control circuitry **16** may supply image content to display **14**. The content may be remotely received (e.g., from a computer or other content source coupled to system **10**) and/or may be generated by control circuitry **16** (e.g., text, other computer-generated content, etc.). The content that is supplied to display **14** by control circuitry **16** may be viewed by a viewer at eye box **24**.

[0025] FIG. **2** is a top view of an illustrative display **14** that may be used in system **10** of FIG. **1**. As shown in FIG. **2**, display **14** may include one or more display projectors such as projector **14A** and an optical system such as optical system **14B**. Optical system **14B** may include optical elements such as one or more waveguides **26**. Waveguide **26** may include one or more stacked substrates (e.g., stacked planar and/or curved layers sometimes referred to herein as waveguide substrates) of optically transparent material such as plastic, polymer, glass, etc.

[0026] If desired, waveguide **26** may also include one or more layers of holographic recording media (sometimes referred to herein as holographic media, grating media, or diffraction grating media) on which one or more diffractive gratings are recorded (e.g., holographic phase gratings, sometimes referred to herein as holograms). A holographic recording may be stored as an optical interference pattern (e.g., alternating regions of different indices of refraction) within a photosensitive optical material such as the holographic media. The optical interference pattern may create a holographic phase grating that, when illuminated with a given light source, diffracts light to create a three-dimensional reconstruction of the holographic recording. The holographic phase grating may be a non-switchable diffractive grating that is encoded with a permanent interference pattern or may be a switchable diffractive grating in which the diffracted light can be modulated by controlling an electric field applied to the holographic recording medium. Multiple holographic phase gratings (holograms) may be recorded within (e.g., superimposed within) the same volume of holographic medium if desired. The holographic phase gratings may be, for example, volume holograms or thin-film holograms in the grating medium. The grating media may include photopolymers, gelatin such as dichromated gelatin, silver halides, holographic polymer dispersed

liquid crystal, or other suitable holographic media.

[0027] Diffractive gratings on waveguide **26** may include holographic phase gratings such as volume holograms or thin-film holograms, meta-gratings, or any other desired diffractive grating structures. The diffractive gratings on waveguide **26** may also include surface relief gratings formed on one or more surfaces of the substrates in waveguides **26**, gratings formed from patterns of metal structures, etc. The diffractive gratings may, for example, include multiple multiplexed gratings (e.g., holograms) that at least partially overlap within the same volume of grating medium (e.g., for diffracting different colors of light and/or light from a range of different input angles at one or more corresponding output angles).

[0028] Optical system **14B** may include collimating optics such as collimating lens **34**. Collimating lens **34** may include one or more lens elements that help direct image light **22** towards waveguide **26**. Collimating lens **34** may be omitted if desired.

[0029] As shown in FIG. **2**, projector **14A** may generate image light **22** associated with image content to be displayed to eye box **24**. Image light **22** may be, for example, light that contains and/or represents something viewable such as a scene or object (e.g., as modulated onto the image light using the image data provided by control circuitry **16** of FIG. **1** to the projector). Image light **22** may, for example, include a stream of image frames (e.g., video data) that contain computer-generated (virtual) objects represented by pixel values in the image frames. Image light **22** may be collimated using a lens such as collimating lens **34**. Optical system **14B** may be used to present image light **22** output from projector **14A** to eye box **24**.

[0030] Optical system **14B** may include one or more optical couplers such as input coupler **28**, cross-coupler **32**, and output coupler **30**. In the example of FIG. **2**, input coupler **28**, cross-coupler **32**, and output coupler **30** are formed at or on waveguide **26**. Input coupler **28**, cross-coupler **32**, and/or output coupler **30** may be completely embedded within the substrate layers of waveguide **26**, may be partially embedded within the substrate layers of waveguide **26**, may be mounted to waveguide **26** (e.g., mounted to an exterior surface of waveguide **26**), etc.

[0031] The example of FIG. **2** is merely illustrative. One or more of these couplers (e.g., cross-coupler **32**) may be omitted. Optical system **14B** may include multiple waveguides that are laterally and/or vertically stacked with respect to each other. Each waveguide may include one, two, all, or none of couplers **28**, **32**, and **30**. Waveguide **26** may be at least partially curved or bent if desired.

[0032] Waveguide **26** may guide image light **22** down its length via total internal reflection. Input coupler **28** may be configured to couple image light **22** from projector **14A** into waveguide **26**, whereas output coupler **30** may be configured to couple image light **22** from within waveguide **26** to the exterior of waveguide **26** and towards eye box **24**. Input coupler **28** may include an input coupling prism, a surface relief grating, louvered mirrors, an angled edge or face of waveguide **26**, volume holograms, metagratings, a reflective layer, and/or other input coupling structures. As an example, projector **14A** may emit image light **22** in the +Y direction towards optical system **14B**. When image light **22** strikes input coupler **28**, input coupler **28** may redirect image light **22** so that the light propagates within waveguide **26** via total internal reflection towards output coupler **30** (e.g., in the +X direction). When image light **22** strikes output coupler **30**, output coupler **30** may redirect image light **22** out of waveguide **26** towards eye box **24** (e.g., back in the -Y direction). In scenarios where cross-coupler **32** is included at waveguide **26**, cross-coupler **32** may redirect image light **22** in one or more directions as it propagates down the length of waveguide **26**, for example. Cross-coupler **32** and/or output coupler **30** may perform one or two dimensional pupil expansion upon redirecting image light **22** if desired.

[0033] Input coupler **28**, cross-coupler **32**, and/or output coupler **30** may be based on reflective and refractive optics or may be based on holographic (e.g., diffractive) optics. In arrangements where couplers **28**, **30**, and **32** are formed from reflective and refractive optics, couplers **28**, **30**, and **32** may include one or more reflectors (e.g., an array of micromirrors, partial mirrors, louvered mirrors, or other reflectors). In arrangements where couplers **28**, **30**, and **32** are based on

holographic optics, couplers **28**, **30**, and **32** may include diffractive gratings (e.g., volume holograms, surface relief gratings, etc.).

[0034] In implementations that are described herein as an example, projector **14A** may include a reflective display panel that generates image light **22** and that provides the image light to waveguide **26**. FIG. **3** is a top view showing how projector **14A** may include a reflective display panel that generates image light **22** for waveguide **26**. As shown in FIG. **3**, projector **14A** may include one or more light sources (LS) **44**. Light sources **44** may be LED light sources, OLED light sources, uLED light sources, laser light sources, and/or any other desired light source.

[0035] Projector **14A** may also include a reflective display panel such as reflective display panel **46** (e.g., a reflective spatial light modulator (SLM)). Reflective display panel **46** may be an LCOS display panel, a DMD display panel, a ferroelectric liquid crystal on silicon (fLCOS) display panel, or another type of reflective display panel. Reflective display panel **46** may have an array of individually adjustable pixels **P**. Each pixel **P** may be formed by a respective reflective element **48** in reflective display panel **46**. Reflective elements **48** are sometimes also referred to herein as reflectors **48** or programable reflectors **48**. In implementations where reflective display panel **46** is a DMD display panel, reflective elements **48** are mirrors such as micromirrors (e.g., micro-electromechanical-systems (MEMS)-based micromirrors), where each mirror forms a respective pixel **P** of reflective display panel **46**.

[0036] Light sources **44** may emit illumination light **40**. Illumination light **40** may include light in one or more wavelength bands (e.g., red, green, and/or blue wavelength bands). For example, light sources **44** may include a first set of one or more light sources that emit a first wavelength range of illumination light **40** (e.g., red wavelengths), a second set of one or more light sources that emit a second wavelength range of illumination light **40** (e.g., green wavelengths), and a third set of one or more light sources that emit a third wavelength range of illumination light **40** (e.g., blue wavelengths).

[0037] Optics **42** in projector **14A** may direct illumination light **40** onto reflective display panel **46**. Optics **42** may include one or more lens elements, prisms, partial reflectors, polarizers, reflective polarizers, filters, X-cubes, and/or other optical components. Optics **42** may, for example, provide illumination light **40** to reflective display panel **46** at an incident angle **B** (relative to the X-axis) across each of the reflective elements **48** in reflective display panel **46**. During operation, control circuitry **16** (FIG. **1**) may control reflective display panel **46** to selectively reflect illumination light **40** at each pixel (reflective element) location to produce image light **22** (e.g., image light having an image as modulated onto the illumination light by the reflective elements **48** in reflective display panel **46**).

[0038] For example, in implementations where reflective display panel **46** is a DMD panel, reflective elements **48** may be individually (e.g., independently) rotatable between two predetermined orientations (states) such as an “ON” state and an “OFF” state. Control circuitry **16** of FIG. **1** may individually adjust the state of each pixel based on the images to be displayed using display **14**. The example of FIG. **3** illustrates the operation of a single pixel **P*** on illumination light **40** for the sake of clarity. However, in general, similar operations are performed at each pixel **P** (reflective element **48**) across the lateral area of reflective display panel **46**. Reflective display panel **46** may include any desired number of pixels arranged into rows and columns or in any other desired pattern (e.g., tens of pixels, hundreds of pixels, thousands of pixels, tens of thousands of pixels, hundreds of thousands of pixels, etc.).

[0039] As shown in FIG. **3**, when pixel **P*** is in the “ON” state, the mirror used to form that pixel **P*** may be at a first orientation (e.g., an “ON” orientation or state). In this orientation, the mirror may reflect illumination light **40** (as image light **22**) at an output angle **A** towards lens **34**. Lens **34** may direct image light **22** towards input coupler **28** of optical system **14B**. When pixel **P*** is in the “OFF” state, pixel **P*** may direct illumination light **40** away from optical system **14B**, as shown by arrow **49** (e.g., towards a light sink **47** such as a baffle that includes light absorbing materials

and/or textured structures that effectively extinguish the reflected light to prevent the reflected light from being received at the eye box). By adjusting pixel P* between the “ON” and “OFF” states in this way, pixel P* may either direct illumination light **40** towards input coupler **28** and the eye box (as image light **22**) or may direct illumination light **40** outside of the projection optics (i.e., towards light sink **47**) so that the beam is not received at the eye box.

[0040] Control circuitry **16** (FIG. **1**) may provide image data DAT to reflective display panel **46** (e.g., over a control bus or path). Image data DAT may specify the state for each pixel P in reflective display panel **46** over time to cause reflective display panel **46** to modulate frames of image data DAT onto illumination light **40**, producing image light **22** having the frames of image data DAT modulated thereon. At any given time, reflective display panel **46** configures its reflective elements **48** according to the corresponding pixel value (e.g., ON or OFF) specified by the image data DAT received by reflective display panel **46**. Control circuitry **16** may, for example, include display rendering circuitry, display pipeline circuitry, and/or display driver circuitry (e.g., as implemented on, controlled by, and/or executed using one or more processors in control circuitry **16**) that generate image data DAT and that provide image data DAT to reflective display panel **46** for modulating a corresponding stream of images onto image light **22**.

[0041] Control circuitry **16** may also control (drive) light sources **44** using a drive signal. The drive signal may include a current signal such as drive current ID and/or a voltage signal such as a drive voltage. Implementations in which the drive signal includes drive current ID are described herein as an example. Control circuitry **16** may adjust the magnitude of the drive signal (e.g., drive current ID) to adjust the intensity of the illumination light **40** produced by each of the light sources **44** in projector **14A** (e.g., where higher signal magnitudes produce higher intensities than lower signal magnitudes). Additionally or alternatively, control circuitry **16** may control the intensity of the illumination light **40** produced by light sources **44** by supplying the drive signal (e.g., drive current ID) to light sources **44** using a corresponding pulse width modulation (PWM) scheme. Control circuitry **16** may adjust the intensity of illumination light **40** by adjusting the PWM scheme (e.g., by adjusting the duty cycle of drive current ID).

[0042] Control circuitry **16** may time-synchronize drive current ID with the image data DAT provided to reflective display panel **46**. When generating image light **22**, control circuitry **16** may supply image data DAT to reflective display panel **46** and may control light sources **44** using a corresponding display modulation sequence. Image data DAT may, for example, include a series of bit planes that are provided, in accordance with the display modulation sequence, to reflective display panel **46** for different durations and synchronized to different colors of the illumination light **40** emitted by light sources **44**.

[0043] FIG. **4** is a timing diagram showing one example of a display modulation sequence **50** that may be used to drive reflective display panel **46**. The bottom row of FIG. **4** describes the state of light sources **44** over time (e.g., the color of the illumination light **40** over time). Blocks **55** graphically represent the wavelength range (color) of the illumination light **40** produced by light sources **44** over time. For example, light sources **44** may generate red (R) illumination light **40** during a first (red) block **55**, may generate green (G) illumination light **40** during a second (green) block **55**, may generate blue (B) illumination light **40** during a third (blue) block **55**, etc. A consecutive set of one red block **55**, one green block **55**, and one blue block **55** is sometimes also referred to herein as a color cycle **60**. Display modulation sequence **50** may include any desired number of color cycles **60** (e.g., a single color cycle **60**, two color cycles **60**, more than two color cycles **60**, etc.) and may include any desired number of blocks **55** of any desired colors in any desired sequence.

[0044] The upper row of FIG. **4** describes the state of reflective display panel **46** as driven using image data DAT. The image data DAT provided to reflective display panel **46** may include a sequence of bit planes **52** over time. Each bit plane **52** contains an array or plane of pixel values (e.g., ON or OFF values), each for a respective one of the pixels P in reflective display panel **46**.

Image data DAT may include image data associated with each color channel of illumination light **40**. For example, bit planes **52** that overlap red blocks **55** include red pixel values and configure reflective display panel **46** to modulate the red pixel values (red image data) onto red illumination light **40** to produce corresponding red image light **22**. Similarly, bit planes **52** that overlap green blocks **55** include green pixel values and configure reflective display panel **46** to modulate the green pixel values (green image data) onto green illumination light **40** to produce corresponding green image light **22**. Similarly, bit planes **52** that overlap blue blocks **55** include blue pixel values and configure reflective display panel **46** to modulate the blue pixel values (blue image data) onto blue illumination light **40** to produce corresponding blue image light **22**. By rapidly cycling through color cycles **60** and the corresponding bit planes **52** faster than the response of the unaided human eye, image light **22** may be perceived as full color image light containing full color images despite the time division duplexing scheme used to modulate image data onto different colors of illumination light **40**.

[0045] The sequence of bit planes **52** overlapping a given block **55** are sometimes also referred to collectively herein as color field **62**. Each color field **62** has an associated color, given by the color of the illumination light **40** illustrated by the overlapping block **55**. Blocks **55** and the overlapping color fields **62** are separated from each other in time by dark periods **56**. Dark periods **56** may accommodate hardware time required by light sources **44** to switch between different colors of illumination light **40** and/or required by reflective display panel **46** to be reconfigured using bit planes **52** of a different color.

[0046] Each bit plane **52** in display modulation sequence **50** has a corresponding bit plane width (duration) **54**, corresponding to the amount of time reflective display panel **46** is configured (driven) using that bit plane **52**. Different bit planes **52** may have different widths **54** or two or more bit planes **52** may have the same width **54**. Suitable setting of the width **54** for each bit plane **52** may effectively weight that bit plane in the generation of image light **22**. Width **54** may sometimes also be referred to herein as bit plane weight **54**. The control circuitry may, for example, increase the width **54** of red bit planes **52** to increase the overall red luminance of image light **22**, may increase the width **54** of blue bit planes **52** to increase the overall blue luminance of image light **22**, may decrease the width **54** of green bit planes **52** to decrease the overall green luminance of image light **22**, etc. The combination of widths **54** across all the bit planes **52** in display modulation sequence **50** may collectively establish the gray level of image light **22**. Bit planes **52** may have a minimum width **54** dictated by the least significant bit (LSB) of the bit planes (e.g., the smallest possible pulse on the digital modulation) given the bit depth of the display modulation sequence. The bit depth is the number of bits used to define each pixel value in the image data. The bits used to define each pixel value have a corresponding LSB. For example, when the bit depth is equal to four, each pixel value is defined by a four-bit number having a most significant bit and a least significant bit. In this example, if a pixel value is “1110,” the least significant bit is “0.”

[0047] When system **10** implements an augmented reality scheme, optical system **14B** transmits light from real-world objects **52** (FIG. **10**) onto eye box **24** and projector **14A** generates image light **22** that includes images of virtual objects. Optical system **14B** combines image light **22** with the light from real-world objects **52** (sometimes also referred to herein as scene light, external light, environmental light, world light, or ambient light), which overlays the images of virtual objects in image light **22** onto the real-world objects viewed through optical system **14B** at eye box **24**.

[0048] Care should be taken to ensure that the images of virtual objects in image light **22** are sufficiently bright so as to be perceivable to the user given the current brightness of the world light. For example, in bright environments, the brightness of projector **14A** and image light **22** may need to be increased to maximize contrast and thus visibility of the images of virtual objects in image light **22**. On the other hand, in dark environments, the brightness of projector **14A** and image light **22** may need to be reduced (dimmed) to avoid user discomfort (e.g., supplying the user with images of virtual objects that are too bright when the user's eyes have otherwise adapted to low levels of

ambient light), to minimize content glare, and to conserve device power.

[0049] If care is not taken, the reflective display panel **46** in projector **14A** can produce undesirable postcard artifacts at eye box **24** that can be distracting to the user in dark environments. Such postcard artifacts are generally perceivable as a slightly visible haze or non-zero brightness filling the entire field of view of eye box **24**, even in portions of the field of view that are not otherwise provided with virtual objects in image light **22**.

[0050] Postcard artifacts have a number of sources associated with the architecture of projector **14A** and optical system **14B**. For example, the entire area of reflective display panel **46** is generally illuminated using illumination light **40** regardless of pixel content or pixel gray level. When reflective display panel **46** is implemented using a DMD panel, stray light from gray level 0 (GL0) pixels (e.g., pixels in the OFF state) can propagate through the system to eye box **24**. When reflective display panel **46** is implemented using an LCOS panel, light from GL0 pixels can also reach the eye box **24** due to leakage from polarization rejection. In addition, some illumination light **40** may produce stray light that reaches eye box **24** via diffraction off of the reflective elements **48** in reflective display panel **46** (e.g., reflective elements **48** may effectively form a diffraction grating that inadvertently diffracts some stray light towards the eye box). Non-idealities in waveguide **26** (FIG. 2) can also contribute to postcard artifacts. For example, surface roughness, spatial variation, line width roughness, and other process variations in one or more diffractive gratings on waveguide **26** can produce stray light that contributes to postcard artifacts. Edge blackening scattering and/or bulk scattering at waveguide **26** can also contribute to postcard artifacts.

[0051] To minimize the visibility of postcard artifacts at eye box **24** at peak virtual content brightness in dark environments, it may be desirable for projector **14A** to reduce its brightness to less than or equal to 5 nits. In some implementations, projector **14A** reduces its brightness using an analog dimming scheme. Under the analog dimming scheme, the drive current ID provided to light sources **44** (FIG. 3) is reduced to minimize the brightness of the illumination light **40** provided to reflective display panel **46**. However, light sources **44** are intrinsically unstable when driven using insufficient current and/or voltage. As such, the drive signal (e.g., drive current ID) can only be reduced to a minimum magnitude before light sources **44** become unstable. Other techniques may be used to further reduce the brightness of image light **22** and projector **14A** without reducing drive current ID below its minimum magnitude.

[0052] In some implementations, the brightness of projector **14A** can be further reduced by reducing the duty cycle of bit planes **52** and/or by inserting one or more dark times (e.g., empty bit planes **52**) into display modulation sequence **50** (e.g., reducing PWM efficiency). However, these techniques may be infeasible because bit planes **52** cannot be duty cycled below the smallest pulse on the digital modulation (e.g., binary PWM). For example, width **54** cannot be reduced to below the LSB of bit planes **52**. To mitigate these issues while minimizing the appearance of postcard artifacts at eye box **24**, projector **14A** may adjust its brightness using a hierarchical brightness adjustment.

[0053] FIG. 5 is a flow chart of illustrative operations involved in adjusting the brightness of projector **14A** while producing image light **22**. At operation **70**, projector **14A** may generate and output image light **22**. Projector **14A** may generate image light **22** by modulating image data DAT onto illumination light **40** (FIG. 3) using one or more display modulation sequences such as display modulation sequence **50** of FIG. 4. Projector **14A** outputs image light **22** at a first display brightness level (e.g., a relatively high display brightness level).

[0054] Processing may proceed to operation **72** in response to a display dimming trigger. The display dimming trigger may be triggered based on sensor data (e.g., ambient light sensor data, camera data, orientation sensor data, etc.) gathered by system **10**, based on a software application running on system **10**, and/or based on a user input received by system **10**. The display dimming trigger may, for example, occur when the image data DAT and the ambient lighting conditions of

system **10** would otherwise cause image light **22** to produce a noticeable postcard artifact at eye box **24** (e.g., when system **10** is operating in a relatively dark environment as identified by ambient light sensor and/or camera sensor data and/or when image light **22** includes relatively bright virtual objects that occupy only a small portion of the field of view of eye box **24**). As another example, the display dimming trigger may occur when a user provides a user input (e.g., a gesture, button press, switch toggle, knob rotation, finger movement, etc.) and/or a software application performs an operation to instruct system **10** to reduce its display brightness. These examples are illustrative and, in general, projector **14A** may perform dimming in response to any desired display dimming trigger. The display dimming trigger may also identify or be associated with a second display brightness level that is lower than the first display brightness level. The second display brightness level may be associated with or given by the current ambient lighting conditions and/or the content to be displayed in image light **22** (e.g., the second display brightness level may be a display brightness required of the projector to sufficiently mitigate postcard artifacts given current ambient light levels and the content to be displayed). A calibration operation may be used to identify the second display brightness level, for example.

[0055] At operation **72** (e.g., responsive to the display dimming trigger), projector **14A** may perform hierarchical display dimming to dim or reduce the brightness of projector **14A** (image light **22**) to the second display brightness level. The hierarchical dimming may smoothly reduce the brightness of image light **22** to levels below what is otherwise achievable by reducing drive current ID to its minimum magnitude. The hierarchical dimming may include adjustments to drive current ID and to the display modulation sequence **50** in the image data DAT provided to reflective display panel **46**. The hierarchical dimming may include timing adjustments (e.g., temporal compression and/or stretching) between display modulation sequences to modify the effective display persistence and to minimize visual artifacts associated with switching between display modulation sequences such as flicker, contouring, dithering artifacts, white point change, etc.

[0056] At operation **74**, projector **14A** may generate and output (dimmed) image light **22** at the second display brightness level. Processing may proceed to operation **76** in response to a display brightening trigger. The display brightening trigger may be triggered based on sensor data (e.g., ambient light sensor data, camera data, orientation sensor data, etc.) gathered by system **10**, based on a software application running on system **10**, and/or based on a user input received by system **10**. The display brightening trigger may, for example, occur when the image data DAT and the ambient lighting conditions of system **10** would not otherwise cause image light **22** to produce noticeable postcard artifacts at eye box **24** (e.g., when system **10** is operating in a relatively bright environment as identified by ambient light sensor and/or camera sensor data and/or when image light **22** includes virtual objects that occupy substantially all of the field of view of eye box **24**). As another example, the display brightening trigger may occur when a user provides a user input (e.g., a gesture, button press, switch toggle, knob rotation, finger movement, etc.) and/or a software application performs an operation to instruct system **10** to increase its display brightness. These examples are illustrative and, in general, projector **14A** may perform brightening in response to any desired display brightening trigger.

[0057] At operation **76** (e.g., responsive to the display brightening trigger), projector **14A** may perform hierarchical display brightening to brighten or increase the brightness of projector **14A** and image light **22** (e.g., back to the first display brightness level or to another display brightness level higher than the second display brightness level). The hierarchical brightening may include reversing the hierarchical dimming performed at operation **72**. The hierarchical brightening may, for example, include adjustments to drive current ID and to the display modulation sequence **50** in the image data DAT provided to reflective display panel **46**. The hierarchical brightening may include timing adjustments (e.g., temporal compression and/or stretching) between display modulation sequences to modify the effective display persistence and to minimize visual artifacts associated with switching between display modulation sequences such as flicker, contouring,

dithering artifacts, white point change, etc. Processing may then loop back to operation **70** via path **78**.

[0058] FIG. **6** is a flow chart of illustrative operations involved in performing hierarchical display dimming using projector **14A** from the first display brightness level to the second display brightness level. The operations of FIG. **6** may, for example, be performed while processing operation **72** of FIG. **5**. The hierarchical display dimming may involve transitioning between three different domains of display dimming as the luminance of the projector is decreased from its maximum or nominal luminance to a luminance associated with the second display brightness level.

[0059] At operation **80** (e.g., in a first domain of display dimming), control circuitry **16** (FIG. **1**) and projector **14A** may perform analog dimming to reduce the luminance (brightness) of image light **22** from a peak luminance to a first threshold luminance **LTHA**. This may include reducing the magnitude of the drive current **ID** provided to light sources **44** (FIG. **3**) and/or adjusting the duty cycle (PWM) of drive current **ID**. Threshold luminance **LTHA** may be the luminance of image light **22** when drive current **ID** is provided to light sources **44** at the minimum magnitude or duty cycle associated with stable operation of light sources **44**.

[0060] If/when the second brightness level corresponds to a luminance greater than threshold luminance **LTHA**, analog dimming is able to sufficiently dim projector **14A** and processing may proceed to operation **74** of FIG. **5** via path **90**. On the other hand, when the second brightness level corresponds to a luminance less than threshold luminance **LTHA**, analog dimming is not sufficient on its own to dim projector **14A** to the second display brightness level given the minimum drive current associated with stable operation of light sources **44**. In these situations, processing proceeds to operation **84** via path **82** to further dim projector **14A**.

[0061] At operation **84** (e.g., in a second domain of display dimming), control circuitry **16** and projector **14A** may perform modulation-based dimming at reflective display panel **46**. If desired, drive current **ID** may be provided to light sources **44** at a constant magnitude and/or duty cycle during modulation-based dimming (e.g., the minimum constant magnitude and/or duty cycle that still allows light sources **44** to emit illumination light **40** without substantial instability). The modulation-based dimming may then allow projector **14A** to further reduce the luminance of image light **22** from the first threshold luminance **LTHA** down to a second threshold luminance

[0062] **LTHB** that is lower than threshold luminance **LTHA**.

[0063] The modulation-based dimming may include adjustment to the display modulation sequences **50** (FIG. **4**) of the image data **DAT** provided to reflective display panel **46**. The adjustment to display modulation sequences **50** may include adjustment to one or more bit planes **52**, one or more color fields **62**, and/or one or more color cycles **60** of display modulation sequence **50** over time (e.g., as a series of display modulation sequences **50** are provided to the reflective display panel in image data **DAT**). The adjustment to display modulation sequences **50** may include reducing the bit depth of subsequent display modulation sequences **50** provided to the reflective display panel in image data **DAT** and/or temporal compression of subsequent display modulation sequences **50** provided to the reflective display panel in image data **DAT**. If desired, the adjustment to display modulation sequences **50** may include temporally stretching one or more of the display modulation sequences to match the luminance produced by the immediately preceding display modulation sequence provided to the reflective display panel in image data **DAT**, helping to mitigate flicker. Similar adjustment may be performed to minimize flicker between performing analog dimming and modulation-based dimming.

[0064] If/when the modulation-based dimming is able to reduce the brightness of projector **14A** and image light **22** to the second display brightness level (e.g., if/when the second brightness level corresponds to a luminance greater than threshold luminance **LTHB**), modulation-based dimming is able to sufficiently dim projector **14A** and processing may proceed to operation **74** of FIG. **5** via path **90**. On the other hand, when the second brightness level corresponds to a luminance less than

threshold luminance LTHB, modulation based dimming and analog dimming are not sufficient on their own to dim projector **14A** to the second display brightness level (e.g., given the LSB of bit planes **52** at a minimum bit depth of the display modulation sequence). In these situations, processing proceeds to operation **88** via path **86** to further dim projector **14A**.

[0065] At operation **88**, (e.g., in a third domain of display dimming), control circuitry **16** and projector **14A** may perform digital (pixel level) dimming at reflective display panel **46**. Rather than adjusting display modulation sequences as in the modulation-based dimming of operation **84**, digital dimming includes dimming or reducing the max gray level of image content (non-zero pixel values) in image data DAT (e.g., at pixels that form virtual objects in the displayed image light **22**). This allows reduction in the brightness of projector **14A** down to the second display brightness level but reduces the contrast of the virtual objects included in image light **22**. Processing may then proceed to operation **74** of FIG. **5**.

[0066] In this way, projector **14A** may first perform simple and efficient analog dimming before switching to modulation-based dimming to further reduce the brightness of the projector below what is otherwise achievable given the minimum current requirements of light sources **44**. Then, if modulation-based dimming is still insufficient to reach the second display brightness level, digital dimming may be performed to trade off contrast for a further reduction in brightness and thus postcard artifacts at eye box **24**.

[0067] The operations of FIG. **6** may be reversed to perform hierarchical display brightening at operation **76** of FIG. **5**. For example, during hierarchical display brightening, projector **14A** may first perform digital (pixel level) brightening in the image data DAT provided to reflective display panel **46**. Then, projector **14A** may perform modulation-based brightening. This may include adjusting display modulation sequences **50** by increasing the bit depth of subsequent display modulation sequences **50** and/or temporal stretching of subsequent display modulation sequences **50** provided to the reflective display panel in image data DAT. If desired, the adjustment to display modulation sequences **50** may include temporally compressing one or more of the display modulation sequences to match the luminance produced by the immediately preceding display modulation sequence provided to the reflective display panel in image data DAT, helping to mitigate flicker. Similar adjustment may be performed to minimize flicker between performing digital brightening and modulation-based brightening and/or between performing modulation-based brightening and analog brightening. If further brightening is required, projector **14A** may then perform analog brightening by increasing the magnitude and/or duty cycle of the drive current ID provided to light sources **44**.

[0068] FIG. **7** is a diagram showing how reflective display panel **46** may perform modulation-based dimming (e.g., while processing operation **84** of FIG. **6**). The vertical axis of FIG. **7** plots the luminance domain of projector **14A**, from a luminance of zero to a luminance exceeding threshold luminance LTHA. Threshold luminance LTHB is greater than zero and less than threshold luminance LTHA. The left column of FIG. **7** illustrates different display modulation sequences **50** that are provided to reflective display panel **46** over time while performing modulation-based dimming. The horizontal dimension of display modulation sequences **50** corresponds to the duration of the display modulation sequence (e.g., the time period during which that display modulation sequence drives reflective display panel **46**). The right column of FIG. **7** illustrates one example of color cycles **62** that may be included in the corresponding display modulation sequences **50** shown in the left column of FIG. **7** (e.g., to help illustrate how bit planes are adjusted between modulation sequences while performing modulation-based dimming).

[0069] In the example of FIG. **7**, hierarchical display dimming is performed to reduce the brightness of projector **14A** from a first display brightness level (FIG. **6**), corresponding to a luminance exceeding threshold luminance LTHA, down to a second display brightness level (FIG. **6**) corresponding to a target luminance LTGT. Analog display dimming (operation **80** of FIG. **6**) is first performed to reduce the brightness of projector **14A** from the first display brightness level to a

brightness level corresponding to threshold luminance LTHA. Modulation-based dimming (operation **84** of FIG. **6**) is then performed to further reduce the brightness of projector **14A** from the brightness level corresponding to threshold luminance LTHA down to a second display brightness level that corresponds to target luminance LTGT. In the example of FIG. **7**, target luminance LTGT exceeds threshold luminance LTHB, so modulation-based dimming is sufficient to reach target luminance LTGT. In examples where target luminance LTGT is less than threshold luminance LTHB, digital dimming (operation **88** of FIG. **6**) is performed after modulation-based dimming to further reduce the luminance of projector **14A** from threshold luminance LTHB down to the target luminance.

[0070] As shown in FIG. **7**, projector **14A** displays image light **22** using an initial display modulation sequence **50-0** at the end of analog dimming (e.g., projector **14A** modulates illumination light **40** using initial display modulation sequence **50-0** while light sources **44** are driven using the minimum stable magnitude and/or duty cycle of drive current ID). Image light **22** has an emission light time (ELT) ELT-1 when produced using initial display modulation sequence **50-0** (e.g., where emission light time is given by integrating over all the bit planes **52** in the display modulation sequence over the duration of the display modulation sequence).

[0071] Projector **14A** may then begin to perform modulation-based dimming. To begin modulation-based dimming, control circuitry **16** (FIG. **1**) may drive reflective display panel **46** (FIG. **3**) using image data DAT that includes a first display modulation sequence **50-1**. Display modulation sequence **50-1** has a lower bit depth than initial display modulation sequence **50-0**. This may, for example, cause display modulation sequence **50-1** to have one or more fewer bit planes **52** than initial display modulation sequence **50-0**. At the same time, the control circuitry may time-stretch display modulation sequence **50-1** to have substantially the same sequence duration **90** as initial display modulation sequence **50-0**, despite its reduced bit depth. This causes reflective display panel **46** to produce, using display modulation sequence **50-1**, image light **22** having approximately the same emission light time ELT-1 as the image light **22** displayed using initial display modulation sequence **50-0**. This may serve to minimize noticeable flickering or other artifacts associated with the reduction in bit depth of display modulation sequence **50-1** relative to initial display modulation sequence **50-0**.

[0072] Block **112** shows one illustrative bit plane configuration for first display modulation sequence **50-1**. In the example of FIG. **7**, as shown by block **112**, first display modulation sequence **50-1** includes nine color fields **62** (e.g., within three color cycles **60** of FIG. **4**). Each color field **62** includes a respective set of bit planes **52** (FIG. **4**) of the corresponding color (not shown in FIG. **7** for the sake of clarity). The cumulative duration of each of the color fields **62** in display modulation sequence **50-1** is equal to sequence duration **90**.

[0073] To further dim image light **22**, as shown by arrow **92**, control circuitry **16** may then drive reflective display panel **46** using image data DAT that includes a time-compressed version of display modulation sequence **50-1**. As shown by block **114**, the time-compressed version of display modulation sequence **50-1** includes the same color fields **62** (as well as the same underlying bit planes) but lasts for a sequence duration **94** that is less than sequence duration **90**. Control circuitry **16** may perform this time compression by reducing the weight (e.g., width **54** of FIG. **4**) of one or more (e.g., all) of the bit planes in display modulation sequence **50-1**. If desired, the ratio of weights/durations of bit planes or color fields of different colors may be maintained both before and after the time compression or may be adjusted upon time compression. Reflective display panel **46** may modulate illumination light **40** using the time-compressed version of display modulation sequence **50-1**, causing the image light to exhibit an emission light time ELT-2 that is less than emission light time ELT-1 (e.g., reducing the luminance of image light **22** relative to before the time-compression of display modulation sequence **50-1**).

[0074] To further dim image light **22**, projector **14A** may continue to time-compress display modulation sequence **50-1** until the width **54** (FIG. **4**) of one or more of the bit planes **52** in display

modulation sequence **50-1** is reduced to a minimum duration **100**. Minimum duration **100** is dictated by the least significant bit (LSB) of display modulation sequence **50-1**, given its bit depth. For example, as shown by arrow **96**, control circuitry **16** may then drive reflective display panel **46** using image data DAT that includes a further time-compressed version of display modulation sequence **50-1**. As shown by block **116**, the further time-compressed version of display modulation sequence **50-1** includes the same color fields **62** (as well as the same underlying bit planes) but lasts for a sequence duration **98** that is even less than sequence duration **94**. Reflective display panel **46** modulates illumination light **40** using the further time-compressed version of display modulation sequence **50-1**, causing the image light to exhibit an emission light time ELT-3 that is less than emission light time ELT-2. In this example, the further time-compressed display modulation sequence **50-1** has at least one bit plane **52** that exhibits minimum duration **100**. As such, projector **14A** will need to adjust the display modulation sequence to perform further dimming.

[0075] To continue modulation-based dimming, control circuitry **16** may then drive reflective display panel **46** using image data DAT that includes a second display modulation sequence **50-2** (as shown by arrow **102**). Display modulation sequence **50-2** has a lower bit depth than display modulation sequence **50-1**. This may, for example, cause display modulation sequence **50-2** to have one or more fewer bit planes **52** than initial display modulation sequence **50-0**. Block **118** shows one illustrative bit plane configuration for display modulation sequence **50-2**. In the example of FIG. 7, as shown by block **118**, display modulation sequence **50-2** includes eight color fields **62** instead of nine as in display modulation sequence **50-1** (e.g., the final blue color field is omitted). This is illustrative and non-limiting. If desired, the reduction in bit depth of display modulation sequence **50-2** relative to display modulation sequence **50-1** may serve to remove any combination of one or more bit planes **52** from one or more of the color fields **62** in the display modulation sequence (e.g., without removing an entire color field), may serve to remove multiple color fields, may serve to remove one or more bit planes from only a single color field/channel, may serve to remove one or more entire color cycles, may serve to remove one or more bit planes from all color fields/channels, etc. If desired, the ratio of weights/durations of bit planes or color fields of different colors may be maintained between display modulation sequences **50-1** and **50-2**.

[0076] At the same time, the control circuitry may time-stretch display modulation sequence **50-2**, to have substantially the same sequence duration **98** as the most recently displayed version of display modulation sequence **50-1** (e.g., the further time compressed version of display modulation sequence **50-1**) despite the reduced bit depth of display modulation sequence **50-2**. This may cause reflective display panel **46** to produce, using display modulation sequence **50-2**, image light **22** having approximately the same emission light time ELT-3 as the image light **22** displayed using the most recent time-compressed version of display modulation sequence **50-1**. This may serve to minimize noticeable flickering or other artifacts associated with the reduction in bit depth of display modulation sequence **50-2** relative to display modulation sequence **50-1**.

[0077] To further dim image light **22**, control circuitry **16** may then continue time-compress display modulation sequence **50-2** until a bit plane **52** in display modulation sequence **50-2** reaches minimum duration **100**. For example, as shown by arrow **104**, control circuitry **16** may then drive reflective display panel **46** using image data DAT that includes a time-compressed version of display modulation sequence **50-2**. As shown by block **120**, the time-compressed version of display modulation sequence **50-2** includes the same color fields **62** (as well as the same underlying bit planes) but lasts for a sequence duration **106** that is even less than sequence duration **98**. Reflective display panel **46** modulates illumination light **40** using the time-compressed version of display modulation sequence **50-2**, causing the image light to exhibit an emission light time ELT-4 that is less than emission light time ELT-3. In this example, the time-compressed display modulation sequence **50-2** has at least one bit plane **52** that exhibits minimum duration **100**. As such, projector **14A** will need to further reduce the bit depth of the display modulation sequence to perform further

dimming.

[0078] Reducing the bit depth of display modulation sequence **50** may create the additional headroom needed to time-stretch each new display modulation sequence to match the duration of the most time-compressed version of the previous display modulation sequence, which serves to minimize flicker. This process of reducing the bit depth of the display modulation sequence, time-compressing the display modulation sequence until the LSB pulse width is reached, and then further reducing the bit depth of the display modulation sequence may continue until image light **22** reaches target luminance LTGT or until image light **22** reaches threshold luminance LTHB. Projector **14A** may switch to digital dimming if/when image light **22** reaches threshold luminance LTHB. In the example of FIG. 7, projector **14A** reaches target luminance LTGT (e.g., corresponding to the second display brightness level of FIG. 5) when a display modulation sequence **50-N** having sequence duration **110** is used to drive reflective display panel **46**. The image light **22** produced by reflective display panel **46** using display modulation sequence **50-N** may reach the eye box with minimal or no visible postcard artifacts.

[0079] This process may be reversed when performing hierarchical display brightening (e.g., at operation **76** of FIG. 5). During modulation-based brightening, projector **14A** may time-stretch and increase the bit depth of the display modulation sequences **50** provided to reflective display panel **46** over time. If desired, projector **14A** may perform time compression upon each increase of bit depth to match the ELT of an increased bit depth display modulation sequence with the most recently used display modulation sequence of lower bit depth. Once threshold luminance LTHA is reached, projector **14A** may perform analog display brightening (e.g., by increasing the magnitude and/or duty cycle of the drive current ID provided to light sources **44**).

[0080] The example of FIG. 7 is illustrative and non-limiting. In general, any desired number of display modulation sequences **50** of different bit depths may be used while performing modulation-based dimming/brightening. If desired, the number of color cycles **60** (FIG. 4) in each display modulation sequence **50** may be adjusted between different display modulation sequences **50** (e.g., between display modulation sequences **50-0** and **50-1**, between display modulation sequences **50-1** and **50-2**, etc.) to allow for further time-stretching of the LSB while preserving the same duty cycle at the transition. As one example, initial display modulation sequence **50-0** may include two color cycles (e.g., having six color fields **62**: RGBRGB) whereas display modulation sequence **50-1** includes only a single color cycle (e.g., having three color fields **62**: RGB), which may provide additional headroom to further stretch the LSB of the sequence while preserving duty cycle. In general, each display modulation sequence **50** may include any desired number of color cycles or fewer than a single color cycle. If desired, projector **14A** may allow a small luminance drop between different display modulation sequences **50** (e.g., by tuning the next display modulation sequence to map a desired brightness ramp profile or brightness ramp step). If desired, analog dimming (e.g., adjustment to drive current ID) and/or digital dimming (e.g., digital pixel value adjustment) may be performed concurrent with modulation-based dimming to reach a desired target luminance (e.g., two or more of operations **80-88** of FIG. 6 may be performed concurrently).

[0081] If desired, each display modulation sequence **50** used during hierarchical dimming or brightening may be calibrated at the illumination source, projector, and full system level (e.g., prior to providing system **10** to an end user or in the field while system **10** is operated by an end user) to enable a seamless transition and to account for variations such as non-idealities in the illumination or driving electronics, as well as part-to-part variations. Such calibration may, for example, involve measuring the optical response of the display when driven under a given modulation sequence and updating the corresponding timing or display modulation settings. In general, the display modulation sequences **50** used during modulation-based dimming/brightening may be optimized to minimize perceptual artifacts while accounting for content brightness level and ambient light levels or environmental conditions. For example, dithering artifacts may be less visible when the brightness of the projector is reduced, allowing a higher tolerance for increasing the LSB temporal

width or reducing the effective bit depth of the transition sequence. Stretch factors for each display modulation sequence **50** may be calibrated to match the luminance at transition relative to the previous display modulation sequence. Sequence design and flicker metrics may be used to optimize the display modulation sequences to account for waveform differences. Native bit depth mismatch may be mitigated by adding additional intermediate display modulation sequences. [0082] FIG. **8** is a flow chart of illustrative operations involved in performing the modulation-based dimming shown in FIG. **7**. The operations of FIG. **8** may, for example, be performed while processing operations **80** and **84** of FIG. **6**.

[0083] At operation **130**, projector **14A** may generate image light **22** using one or more display modulation sequences **50** while reducing the luminance of illumination light **40** (e.g., while performing analog dimming at operation **80** of FIG. **6**). The final display modulation sequence used while performing analog dimming may be display modulation sequence **50-0** of FIG. **8**, for example. Processing proceeds to operation **132** when the minimum magnitude and/or duty cycle of the drive current **ID** supplied to light sources **44** is reached.

[0084] At operation **132**, control circuitry **16** (FIG. **1**) may generate a first display modulation sequence with a lower bit depth than initial display modulation sequence **50-0** (e.g., display modulation sequence **50-1** of FIG. **7**). Control circuitry **16** may also time-stretch the first display modulation sequence to exhibit a sequence duration **90** that matches the sequence duration of initial display modulation sequence **50-0** (e.g., to mitigate flicker between use of display modulation sequences **50-0** and **50-1**).

[0085] At operation **134**, projector **14A** may generate image light **22** by modulating illumination light **40** using the first display modulation sequence.

[0086] At operation **136**, projector **14A** may continue to generate image light **22** by modulating illumination light **40** using one or more time-compressed versions of the first display modulation sequence (e.g., may generate image light **22** at a first time using the time-compressed display modulation sequence **50-1** shown by block **114** of FIG. **7**, may generate image light **22** at a second time using the further time-compressed display modulation sequence **50-1** shown by block **116** of FIG. **7**, etc.). If/when projector **14A** reaches target luminance **LTGT** (or the second display brightness level of FIG. **5**) before the time-compressed versions of display modulation sequence **50-1** have a bit plane **52** with a width **54** that is less than or equal to minimum duration **100**, processing may proceed to operation **74** of FIG. **5** via path **138**. On the other hand, if/when time-compressed display modulation sequence **50-1** has a bit plane **52** with a width **54** that is less than or equal to minimum duration **100**, processing may proceed to operation **142** via path **140**.

[0087] At operation **142**, control circuitry **16** may generate an updated display modulation sequence (e.g., display modulation sequence **50-2** of FIG. **7**) having a lower bit depth than the previous display modulation sequence used to generate image light **22**. Control circuitry **16** may also time-stretch the updated display modulation sequence to exhibit a sequence duration that matches the sequence duration of the most time-compressed version of the previous display modulation sequence used to produce image light **22** (e.g., to mitigate flicker between use of display modulation sequence **50-2** and the most time-compressed version of display modulation sequence **50-1**).

[0088] At operation **144**, projector **14A** may begin to generate image light **22** by modulating illumination light **40** using the updated display modulation sequence (e.g., display modulation sequence **50-2** of FIG. **7**). Projector **14A** may continue to generate image light **22** by time-compressing the updated display modulation sequence until the updated display modulation sequence has a bit plane **52** with a width **54** that is less than or equal to minimum duration **100**. If/when the time-compressed versions of the updated display modulation sequence have a bit plane with a width **54** that is less than or equal to minimum duration **100** (e.g., a bit plane that reaches the **LSB** pulse width) and the luminance image light **22** exceeds threshold luminance **LTHB**, processing may loop back to operation **142** via path **150** and a new updated display modulation

sequence having an even lower bit depth may be used. If/when projector **14A** reaches target luminance LTGT (or the second display brightness level of FIG. **5**) before the time-compressed versions of the updated display modulation sequence have a bit plane **52** with a width **54** that is less than or equal to minimum duration **100**, processing may proceed to operation **74** of FIG. **5** via path **146** (e.g., when display modulation sequence **50-N** of FIG. **7** is used, causing the projector to produce image light **22** at target luminance LTGT). On the other hand, if/when projector **14A** produces image light **22** having a luminance less than or equal to threshold LTHB, modulation-based dimming may be insufficient to reach the second display brightness level and processing may proceed to digital dimming (e.g., to operation **88** of FIG. **6** via path **148**). The operations of FIG. **8** may be time-reversed to perform modulation-based brightening (e.g., while processing operation **76** of FIG. **5**).

[0089] The methods and operations described above in connection with FIGS. **1-8** may be performed by the components of system **10** using software, firmware, and/or hardware (e.g., dedicated circuitry or hardware). Software code for performing these operations may be stored on non-transitory computer readable storage media (e.g., tangible computer readable storage media) stored on one or more of the components of system **10** (e.g., the storage circuitry within control circuitry **16** of FIG. **1**). The software code may sometimes be referred to as software, data, instructions, program instructions, or code. The non-transitory computer readable storage media may include drives, non-volatile memory such as non-volatile random-access memory (NVRAM), removable flash drives or other removable media, other types of random-access memory, etc. Software stored on the non-transitory computer readable storage media may be executed by processing circuitry on one or more of the components of system **10** (e.g., one or more processors in control circuitry **16**). The processing circuitry may include microprocessors, application processors, digital signal processors, central processing units (CPUs), application-specific integrated circuits with processing circuitry, or other processing circuitry.

[0090] As used herein, the term “concurrent” means at least partially overlapping in time. In other words, first and second events are referred to herein as being “concurrent” with each other if at least some of the first event occurs at the same time as at least some of the second event (e.g., if at least some of the first event occurs during, while, or when at least some of the second event occurs). First and second events can be concurrent if the first and second events are simultaneous (e.g., if the entire duration of the first event overlaps the entire duration of the second event in time) but can also be concurrent if the first and second events are non-simultaneous (e.g., if the first event starts before or after the start of the second event, if the first event ends before or after the end of the second event, or if the first and second events are partially non-overlapping in time). As used herein, the term “while” is synonymous with “concurrent.” The term “when” also implies at least some concurrency (e.g., event A occurring “when” event B occurs means that at least some of event A is concurrent with at least some of event B).

[0091] System **10** may gather and/or use personally identifiable information. It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

[0092] Physical environment: A physical environment refers to a physical world that people can sense and/or interact with without aid of electronic systems. Physical environments, such as a physical park, include physical articles, such as physical trees, physical buildings, and physical people. People can directly sense and/or interact with the physical environment, such as through sight, touch, hearing, taste, and smell.

[0093] Computer-generated reality: in contrast, a computer-generated reality (CGR) environment refers to a wholly or partially simulated environment that people sense and/or interact with via an

electronic system. In CGR, a subset of a person's physical motions, or representations thereof, are tracked, and, in response, one or more characteristics of one or more virtual objects simulated in the CGR environment are adjusted in a manner that comports with at least one law of physics. For example, a CGR system may detect a person's head turning and, in response, adjust graphical content and an acoustic field presented to the person in a manner similar to how such views and sounds would change in a physical environment. In some situations (e.g., for accessibility reasons), adjustments to characteristic(s) of virtual object(s) in a CGR environment may be made in response to representations of physical motions (e.g., vocal commands). A person may sense and/or interact with a CGR object using any one of their senses, including sight, sound, touch, taste, and smell. For example, a person may sense and/or interact with audio objects that create 3D or spatial audio environment that provides the perception of point audio sources in 3D space. In another example, audio objects may enable audio transparency, which selectively incorporates ambient sounds from the physical environment with or without computer-generated audio. In some CGR environments, a person may sense and/or interact only with audio objects. Examples of CGR include virtual reality and mixed reality.

[0094] Virtual reality: A virtual reality (VR) environment refers to a simulated environment that is designed to be based entirely on computer-generated sensory inputs for one or more senses. A VR environment comprises a plurality of virtual objects with which a person may sense and/or interact. For example, computer-generated imagery of trees, buildings, and avatars representing people are examples of virtual objects. A person may sense and/or interact with virtual objects in the VR environment through a simulation of the person's presence within the computer-generated environment, and/or through a simulation of a subset of the person's physical movements within the computer-generated environment.

[0095] Mixed reality: In contrast to a VR environment, which is designed to be based entirely on computer-generated sensory inputs, a mixed reality (MR) environment refers to a simulated environment that is designed to incorporate sensory inputs from the physical environment, or a representation thereof, in addition to including computer-generated sensory inputs (e.g., virtual objects). On a virtuality continuum, a mixed reality environment is anywhere between, but not including, a wholly physical environment at one end and virtual reality environment at the other end. In some MR environments, computer-generated sensory inputs may respond to changes in sensory inputs from the physical environment. Also, some electronic systems for presenting an MR environment may track location and/or orientation with respect to the physical environment to enable virtual objects to interact with real objects (that is, physical articles from the physical environment or representations thereof). For example, a system may account for movements so that a virtual tree appears stationery with respect to the physical ground. Examples of mixed realities include augmented reality and augmented virtuality. Augmented reality: an augmented reality (AR) environment refers to a simulated environment in which one or more virtual objects are superimposed over a physical environment, or a representation thereof. For example, an electronic system for presenting an AR environment may have a transparent or translucent display through which a person may directly view the physical environment. The system may be configured to present virtual objects on the transparent or translucent display, so that a person, using the system, perceives the virtual objects superimposed over the physical environment. Alternatively, a system may have an opaque display and one or more imaging sensors that capture images or video of the physical environment, which are representations of the physical environment. The system composites the images or video with virtual objects, and presents the composition on the opaque display. A person, using the system, indirectly views the physical environment by way of the images or video of the physical environment, and perceives the virtual objects superimposed over the physical environment. As used herein, a video of the physical environment shown on an opaque display is called "pass-through video," meaning a system uses one or more image sensor(s) to capture images of the physical environment, and uses those images in presenting the AR

environment on the opaque display. Further alternatively, a system may have a projection system that projects virtual objects into the physical environment, for example, as a hologram or on a physical surface, so that a person, using the system, perceives the virtual objects superimposed over the physical environment. An augmented reality environment also refers to a simulated environment in which a representation of a physical environment is transformed by computer-generated sensory information. For example, in providing pass-through video, a system may transform one or more sensor images to impose a select perspective (e.g., viewpoint) different than the perspective captured by the imaging sensors. As another example, a representation of a physical environment may be transformed by graphically modifying (e.g., enlarging) portions thereof, such that the modified portion may be representative but not photorealistic versions of the originally captured images. As a further example, a representation of a physical environment may be transformed by graphically eliminating or obfuscating portions thereof. Augmented virtuality: an augmented virtuality (AV) environment refers to a simulated environment in which a virtual or computer generated environment incorporates one or more sensory inputs from the physical environment. The sensory inputs may be representations of one or more characteristics of the physical environment. For example, an AV park may have virtual trees and virtual buildings, but people with faces photorealistically reproduced from images taken of physical people. As another example, a virtual object may adopt a shape or color of a physical article imaged by one or more imaging sensors. As a further example, a virtual object may adopt shadows consistent with the position of the sun in the physical environment.

[0096] Hardware: there are many different types of electronic systems that enable a person to sense and/or interact with various CGR environments. Examples include head mounted systems, projection-based systems, heads-up displays (HUDs), vehicle windshields having integrated display capability, windows having integrated display capability, displays formed as lenses designed to be placed on a person's eyes (e.g., similar to contact lenses), headphones/earphones, speaker arrays, input systems (e.g., wearable or handheld controllers with or without haptic feedback), smartphones, tablets, and desktop/laptop computers. A head mounted system may have one or more speaker(s) and an integrated opaque display. Alternatively, a head mounted system may be configured to accept an external opaque display (e.g., a smartphone). The head mounted system may incorporate one or more imaging sensors to capture images or video of the physical environment, and/or one or more microphones to capture audio of the physical environment. Rather than an opaque display, a head mounted system may have a transparent or translucent display. The transparent or translucent display may have a medium through which light representative of images is directed to a person's eyes. The display may utilize digital light projection, OLEDs, LEDs, μ LEDs, liquid crystal on silicon, laser scanning light sources, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In one embodiment, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface.

[0097] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. An electronic device comprising: one or more light sources configured to generate illumination light based on a drive signal; a reflective display panel configured to receive image data and configured to generate image light by reflecting the illumination light based on the image data; a

waveguide configured to direct the image light towards an eye box; and one or more processors configured to adjust the image light between a first brightness and a second brightness using the drive signal, the second brightness being lower than the first brightness, and adjust the image light between the second brightness and a third brightness by adjusting bit planes of a display modulation sequence in the image data, the third brightness being lower than the second brightness.

2. The electronic device of claim 1, the one or more processors being further configured to: adjust the image light between the third brightness and a fourth brightness by digitally adjusting pixel values of the image data, the fourth brightness being lower than the third brightness.
3. The electronic device of claim 1, the one or more processors being further configured to: supply the drive signal to the one or more light sources at a first magnitude when the image light is at the first brightness, supply the drive signal to the one or more light sources at a second magnitude lower than the first magnitude when the image light is at the second brightness, and supply the drive signal to the one or more light sources at the second magnitude when the image light is between the second brightness and the third brightness.
4. The electronic device of claim 1, the one or more processors being further configured to: supply the drive signal to the one or more light sources using a first pulse width modulation when the image light is at the first brightness, supply the drive signal to the one or more light sources using a second pulse width modulation different from the first pulse width modulation when the image light is at the second brightness, and supply the drive signal to the one or more light sources using the second pulse width modulation when the image light is between the second brightness and the third brightness.
5. The electronic device of claim 1, the one or more processors being further configured to adjust the image light between the second brightness and the third brightness by adjusting a bit depth of the display modulation sequence.
6. The electronic device of claim 5, wherein the bit depth of the display modulation sequence is higher when the image light is at a higher brightness between the second brightness and the third brightness than when the image light is at a lower brightness between the second brightness and the third brightness.
7. The electronic device of claim 6, the one or more processors being further configured to adjust the image light between the second brightness and the third brightness by adjusting a duration of the display modulation sequence.
8. The electronic device of claim 7, wherein the duration of the display modulation sequence is higher when the image light is at the higher brightness between the second brightness and the third brightness than when the image light is at the lower brightness between the second brightness and the third brightness.
9. The electronic device of claim 1, the one or more processors being further configured to adjust the image light between the second brightness and the third brightness by adjusting a duration of the display modulation sequence.
10. The electronic device of claim 9, wherein the duration of the display modulation sequence is higher when the image light is at a higher brightness between the second brightness and the third brightness than when the image light is at a lower brightness between the second brightness and the third brightness.
11. An electronic device comprising: one or more light sources configured to generate illumination light; a reflective display panel configured to receive image data and configured to generate image light by reflecting the illumination light based on the image data, wherein the image data implements a display modulation sequence; a waveguide configured to direct the image light towards an eye box; and one or more processors configured to dim the image light by at least reducing a bit depth of the display modulation sequence.
12. The electronic device of claim 11, the one or more processors being further configured to dim the light by time-compressing the display modulation sequence.

- 13.** The electronic device of claim 11, the one or more processors being further configured to: drive, at a first time, the reflective display panel using a first display modulation sequence having a first bit depth; and drive, at a second time after the first time, the reflective display panel using a second display modulation sequence having a second bit depth less than the first bit depth.
- 14.** The electronic device of claim 13, the one or more processors being further configured to: drive, at a third time after the second time, the reflective display panel using a third display modulation sequence having a third bit depth less than the second bit depth, wherein the image light has a first brightness while the reflective display panel is driven using the first display modulation sequence and has a second brightness less than the first brightness while the reflective display panel is driven using the third display modulation sequence.
- 15.** The electronic device of claim 14, wherein the first display modulation sequence has a first duration, the second display modulation sequence has the first duration, and the third modulation sequence has a second duration less than the first duration.
- 16.** The electronic device of claim 13, the one or more processors being further configured to: drive, at a third time after the second time, the reflective display panel using a time-compressed version of the second display modulation sequence.
- 17.** A method of operating an electronic device, the method comprising: generating, using one or more light sources, illumination light; generating, using a reflective display panel, image light based on the illumination light; directing, using a waveguide, the image light towards an eye box; and dimming, using one or more processors, the image light, wherein dimming the image light comprises driving, at a first time, the reflective display panel using a first sequence of bit planes for a first duration, and driving, at a second time after the first time, the reflective display panel using the first sequence of bit planes for a second duration less than the first duration.
- 18.** The method of claim 17, wherein dimming the image light further comprises: driving, at a third time after the second time, the reflective display panel using a second sequence of bit planes, the second sequence having lower bit depth than the first sequence.
- 19.** The method of claim 18, wherein driving the reflective display panel using the second sequence of bit planes comprises driving the reflective display panel using the second sequence of bit planes for the second duration.
- 20.** The method of claim 19, dimming the image light further comprises: driving, at a fourth time after the third time, the reflective display panel using a third sequence of bit planes, the third sequence having lower bit depth than the second sequence.
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